

DO NOT WRITE ON TEST



**FINANCIAL MATH &
ANALYSIS CONCEPTS**

~OPEN EVENT~

(190)

NATIONAL 2022

TOTAL POINTS

_____100 Points

Test Time: 60 minutes

MULTIPLE CHOICE

Identify the choice that *best* completes the statement or answers the question.

1. Neil deducts \$250 from his paycheck each month for his insurance premiums. His overall tax rate is 16.25%. If Neil has an annual salary of \$93,000, what will Neil's monthly net pay be?
 - a. \$6,490.63
 - b. \$6,281.25
 - c. \$6,240.63
 - d. \$6,473.18
2. Jan gets paid every two weeks. She worked 46 hours one week and 39 hours the next week. Her hourly pay rate is \$11.30. If taxes are 13%, how much is Jan's paycheck?
 - a. \$994.40
 - b. \$960.50
 - c. \$835.64
 - d. \$865.13
3. Theo is considering switching jobs. He is looking at how each job pays. His current job pays monthly, but his potential job pays every two weeks. If his monthly gross pay is \$3333 and his annual earnings won't change, how much will his new gross pay be?
 - a. \$1666.50
 - b. \$1538.31
 - c. \$1461.39
 - d. \$1583.18
4. Donna was just promoted to a sales manager. As manager, her job is to support her sales staff and as incentive, she receives a 1% commission of her team's sales. What would Donna's commission be if her team members had sales of \$223,678; \$351,190; \$417,732; \$276,333; and \$198,288?
 - a. \$16,139.43
 - b. \$13,204.99
 - c. \$14,275.21
 - d. \$14,672.21
5. Papa's Autos pays its sales associates a monthly base pay of \$1,500. If associates meet a \$50,000 threshold of sales, they will also earn a 5% commission on their sales. If an employee of Papa's Autos sells \$72,366 worth of cars, what would their pay be?
 - a. \$2,118.30
 - b. \$3,618.30
 - c. \$5,118.30
 - d. \$6,618.30

6. Priscilla works for a home cleaning service. She makes 25% commission on the homes she cleans, but her company offers her 50% commission on homes that Priscilla contracts herself. Typically, home cleanings are \$1,000 per home. If Priscilla cleaned 12 homes last week and contracted four of those herself, how much commission did she earn?
- \$4,000
 - \$3,000
 - \$5,000
 - \$6,000
7. Kelsie bought a new office phone. She borrowed \$1,200 at a rate of 18% for 9 months. How much interest will she pay?
- \$216.00
 - \$162.00
 - \$121.50
 - \$270.00
8. Jean owes \$38,000 in student loans for college. The interest rate is 6.25% and the loan will be paid off over 10 years. How much will Jan pay to pay off the loan?
- \$23,750
 - \$65,550
 - \$61,750
 - \$27,550
9. You are starting your own small business in Albuquerque. You borrow \$10,000 from the bank at a 9% rate for five years. How much interest will you pay on the loan?
- \$4,500
 - \$4,750
 - \$4,250
 - \$4,333
10. Easy Gutter, Inc. sells and installs rain gutter on homes and commercial properties. Recently, their daily installations have been 3, 4, 5.5, 4.5, 4, 4, 2, 3, 2.5, 4.5. The daily average is 3.7 installations. What is the median and mode?
- median=4; mode=4
 - median=3.5; mode=4
 - median=3.5; mode=4.5
 - median=4; mode=4.5
11. Calm Cool Refrigeration has three sales associates. Their associates have an annual sales average of \$354,164. Which of the following could be the sales numbers for the associates?
- \$251,453; \$515,734; \$633,998
 - \$401,324; \$454,834; \$376,886
 - \$325,794; \$333,845; \$289,107
 - \$279,474; \$443,290; \$339,728

12. Soda Pop Sweets manufactures many popular candies that taste like soda flavors. The weekly usage of sugar to make their products last month were (measured in tons) 114, 145, 132, and 127. On average, how much sugar (in tons) should Soda Pop Sweets plan to have on hand to complete their manufacturing each week?
- 145
 - 129.5
 - 127
 - 134.5
13. Street Sweepers has an annual revenue of \$1.4 million and they have eight employees. What is the sales per employee ratio?
- 5.71 million
 - .175 million
 - .751 million
 - 7.51 million
14. Capital Financial has 2.4 million in liabilities and 5.4 million in assets. What is the ratio of assets to liabilities?
- 12 to 27
 - 27 to 12
 - 2.75 to 1
 - 1 to 2.75
15. Richard decided to make his office space 40% larger. The current dimensions are 12 feet by 14 feet. Which of the following dimensions match the 40% increase?
- 12 ft x 14 ft
 - 14 ft x 16.8 ft
 - 16.8 ft x 19.6 ft
 - 12 ft x 21.9 ft
16. What is $\frac{3}{8}$ of 100?
- 33.33
 - 18.75
 - 12.5
 - 37.5
17. Talks-a-lot has a sale for phones. If two are purchased, you get $\frac{1}{3}$ off the price. If you buy two phones at \$1,299 each, what would the price be after the discount?
- \$2,598
 - \$2,132
 - \$1,948
 - \$1,732

18. Charlie found that for about every \$2 he put into his business, he could get \$5 in return. If he put \$32,500 into his business, what could he expect to gain back?
- a. \$162,500
 - b. \$97,500
 - c. \$81,250
 - d. \$65,000
19. When providing a quote for a new house, Sore Sight Construction company quotes \$1,500 per 10 square feet and \$25 per foot of fence. If they are asked to build a 1,200 square foot house with 100 feet of fence, how much should they quote?
- a. \$180,000
 - b. \$182,500
 - c. \$185,000
 - d. \$187,500
20. Cozy Home sells appliances and typically sells a set with four appliances per set. Last month, they sold 310 appliances and made \$650,000. Based on this information, how much would one set cost?
- a. \$8,387.10
 - b. \$6,290.32
 - c. \$10,483.87
 - d. \$2,096.77
21. Justine went to the local superstore and bought \$82.50 in electronics, \$41.10 in clothes, \$16.10 in candy, and \$119.22 in food. The sales tax for the local area is 8.5%. What would the total cost be?
- a. \$280.93
 - b. \$280.92
 - c. \$279.63
 - d. \$279.64
22. If an invoice has terms of net 60 and is dated January 16, what day must the invoice be paid in full?
- a. March 16
 - b. March 17
 - c. March 18
 - d. March 19

23. Eye Catching Graphics bought supplies for \$24,000 before tax which is 5%. The terms of the invoice are 2/15 net 30. What will be the final amount of the invoice if paid 13 days after the invoice date?
- \$25,200
 - \$24,120
 - \$24,696
 - \$23,969
24. Jerry's accounting firm received an invoice for \$31,250 and terms of 2/10 net 45. If paid 12 days after the invoice date, how much is due?
- \$30,312.50
 - \$30,626.00
 - \$30,937.50
 - \$31,250.00
25. A leather jacket can be bought from the wholesaler for \$65. The price at a retail store is \$113.75. What is the markup rate?
- 75%
 - 72.5%
 - 57.1%
 - 55.7%
26. An auto parts dealer pays \$7.14 per 12 quarts of oil and the markup is 50% on selling price. Find the selling price per quart.
- \$14.28
 - \$15.36
 - \$1.19
 - \$2.38
27. From the manufacturer, you can buy a cell phone case for \$6.50. The wholesaler will buy it and mark it up 60%. Local phone stores will buy the phone case from the wholesaler and mark it up 45%. How much would the local phone store sell the case for?
- \$7.17
 - \$15.08
 - \$11.49
 - \$13.37
28. The cost for 36 items is \$540. If six of the items cannot be sold, what is the selling price per item for a 25% markup?
- \$21.00
 - \$22.50
 - \$24.00
 - \$25.50

29. Mario's Cakes can make 300 plain cheesecakes at \$2.25 each. If they experience a spoilage rate of 3%, what price should each cheesecake be sold at in order to have a 30% markup?
- \$2.76
 - \$2.87
 - \$3.13
 - \$3.02
30. Fruity Concoctions buys fruit for gift baskets. If they bought 150 lbs. of apples purchased at \$0.75 a lb. and like to have a markup of 40%, how much should they charge per lb. of apples if they expect to have 8% spoil?
- \$1.14
 - \$1.25
 - \$1.31
 - \$1.05
31. Kitchen's Unlimited purchased 7 refrigerators at \$1000; 6 more at \$1025; and finally, 7 at \$980. They were able to sell 8 units. If they use the FIFO method, what would they record as the cost of the sold units?
- \$7,885
 - \$8,150
 - \$8,025
 - \$8,110
32. Corner Store Gas Station bought 10 candy bars at \$2.35 and then another 10 at \$2.45. They have sold 9 candy bars. Using the FIFO method, what is the cost of the inventory left on hand?
- \$26.85
 - \$25.95
 - \$26.95
 - \$25.85
33. Office Supplies, Inc. bought some new planners. They had 4 shipments of 20 planners. The planners in the shipments cost \$11.00 in the first one, \$10.50 in the second one, \$10.65 in the third one, and \$10.80 in the final shipment. Later, they sold 25 planners. Using the LIFO method, how much did the 25 sold planners cost?
- \$272.50
 - \$269.25
 - \$267.00
 - \$265.00

34. Tommy's Supplies uses the weighted average inventory method. They purchased 10 jackets at \$100 and then 10 more at \$90. They sold 5. How much do the remaining jackets cost?
- \$1,400
 - \$1,500
 - \$1,425
 - \$1,450
35. Shannon purchased a sewing machine for \$1375. When purchasing, the sales agent mentioned the machine should last for about 12 years and loses about \$100 of value a year. Sticking with a straight-line depreciation method, what is the salvage value of the machine?
- \$115
 - \$75
 - \$100
 - \$175
36. When using the double-declining balance depreciation method, what is the rate of depreciation if the useful life is 6 years?
- 6
 - $\frac{1}{6}$
 - $\frac{1}{3}$
 - .5
37. Mr. Danielson bought a \$6,000 printer for his high school classroom to use as a learning tool and a fundraiser for his program. The printer he bought has a depreciation rate of 30%, salvage value of \$500, and a useful life of 7 years. What would be the depreciation expense on the printer for the first year if Mr. Danielson's school is using a declining balance method?
- \$1,800
 - \$2,000
 - \$1,155
 - \$1,650
38. An asset valuing \$100,000 with a four-year life and no salvage value would have how much depreciation in the first year using the double-declining balance method?
- \$25,000
 - \$50,000
 - \$20,000
 - \$45,000
39. Which depreciation method lowers the value of assets the fastest?
- straight-line
 - sum-of-the-years' digits
 - double-declining balance
 - declining balance

40. Home Construction offers a trade discount of 5% on orders over \$10,000 and a cash discount of 3% if paid within 30 days. A customer has an invoice of \$36,373 and paid it at 15 days. What did this customer pay Home Construction?
- \$33,463.16
 - \$35,281.81
 - \$34,554.35
 - \$33,517.72
41. All-In-One-Place Wholesale offers a trade discount on orders with an overall quantity of 10,000 items or more. The discount is 5%. A customer orders 14,500 of an item costing \$56.75 each. How much is the discount?
- \$24,686.25
 - \$32,915.00
 - \$41,143.75
 - \$49,372.50
42. Z's Mattresses allow customers to save 4% in a cash discount if they pay their total amount within 45 days. If a customer owes \$1,230.44 and pays half of it at day 40, and the other half at day 50, how much will this customer end up paying?
- \$1,230.44
 - \$1,205.83
 - \$1,181.22
 - \$1,158.73
43. Dan has two CD's each with \$1,000 deposited in them. One has 6% interest compounded bi-monthly. The second has 6% interest compounded semiannually. What is the value of the both CD's combined in two years?
- \$2,480.00
 - \$2,486.83
 - \$2,245.51
 - \$2,252.34
44. Your credit union has advertised two different accounts. One is a CD with 1% interest compounded annually. The other is a savings account with 1% compounded monthly. If you have \$600, which account will give you the highest future value for your money in one year and how much will that value be?
- CD higher with value of \$606.00
 - CD higher with value of \$606.03
 - Savings higher with value of \$606.03
 - Savings higher with value of \$606.00

45. Molly decided to invest some extra money she has in the stock market (\$750). Current trends show that money in the stock market grows at 3% each month. If that trend continues, what will the future value of Molly's money be in 2 months?
- \$795.00
 - \$795.68
 - \$772.50
 - \$773.00
46. Peter has \$120,000. He would like to invest it, but only if he can get a return greater than \$10,000 per year. What is the minimum percent interest would he need to get that amount?
- 8.0%
 - 8.5%
 - 9.0%
 - 9.5%
47. John buys 100 shares of stock at \$75 per share. The price goes up by 10% and he sells 50 shares. Then, prices drop by 8% and he sells his remaining 50 shares. How much did he get for the last 50?
- \$3,795
 - \$3,450
 - \$4,600
 - \$2,250
48. Olivia Inc. uses the allowance method to estimate bad debts. On January 1, Olivia had Accounts Receivable of \$64,000 and Allowance for Bad Debts had a credit balance of \$4,200. During the year, Olivia had sales of \$157,000 (\$37,000 was in cash), and collections on account of \$97,000. Write-offs for the year were \$5,700. What is Olivia's balance in Accounts Receivable on December 31?
- \$118,300
 - \$114,100
 - \$77,100
 - \$81,300
49. Saunders Co. purchased new office space for \$320,000. They put \$25,000 down and signed a 30-year mortgage at 3.75% for the rest. The monthly payments are \$1,367. How much of the first payment will be applied towards the principal of the loan?
- \$367.00
 - \$1,000.00
 - \$921.88
 - \$445.13
50. Schumacher Inc. issued \$100,000 5-year 8% bonds that sold at 1.0374. How much cash did Schumacher receive for this transaction?
- \$100,000
 - \$103,740
 - \$108,000
 - \$96,260